

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	35.0	Gender	Male
Height (cm)	162.0	Weight (kg)	85.0
BMI	32.39	Product Info 2	D3
Occupation	prof-specialty	Annual Income	\$50,000.00
Insurance Type	Health	Coverage Amount	\$650,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 3 of 8
Risk Category	Medium Risk
Recommended Annual Premium	\$14,000.00

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

Application ID: officer@shield.com

Date: October 26, 2023

1. Risk Assessment

The applicant presents a mixed risk profile.

* **Age:** 35.0 - Moderate risk factor. Generally, health risks begin to escalate significantly after this age, so this is a favorable age.

* **BMI:** 32.39 (Height: 162cm, Weight: 85kg) - **High Risk Factor**. A BMI over 30 is classified as obese, which significantly increases the risk for various health conditions (e.g., cardiovascular disease, diabetes, musculoskeletal issues) and is a primary concern for Health insurance.

* **Smoking Status:** 0 (Non-smoker) - **Favorable Risk Factor**. Eliminates a major health hazard.

* **Occupation:** 'prof-specialty' - Generally considered a **Favorable Risk Factor** with lower physical hazard and potentially stable income, though specific job duties are not detailed.

* **Income:** \$50,000.00 - Moderate income level.

* **Previous Claims:** 0 - **Favorable Risk Factor**. Indicates no history of claims.

* **Family History:** 0 - **Favorable Risk Factor**. Suggests no known hereditary predispositions to major illnesses.

* **Exercise:** 1 (Yes) - **Favorable Risk Factor**. Mitigates some health risks, potentially offsetting the BMI to some extent.

* **Alcohol:** 0 (No) - **Favorable Risk Factor**. Eliminates alcohol-related health risks.

* **Gender:** Male - Standard risk factor.

* **Insurance Type:** Health - Focuses risk assessment heavily on current and potential health conditions.

* **Coverage Amount:** \$650,000.00 - **High coverage for health insurance.**

Overall: The most significant concern is the applicant's obesity (BMI 32.39). This is somewhat offset by positive lifestyle factors (non-smoker, exercises, no alcohol) and a clean history (no previous claims, no family history).

2. Comparative Analysis

The current application (Predicted Risk Class: 3, Premium: \$14000.00) sits within the range of the provided similar historical approved cases.

* **Application POL-2626 (Similarity: 42.8%):** This case had a lower Risk Class (2) and a lower Premium (\$12000.00). The current applicant's higher BMI likely contributes to the slightly higher risk class and premium compared to POL-2626, despite similar positive lifestyle factors.

* **Application POL-25755 (Similarity: 40.0%):** This case had a significantly higher Risk Class (5) and Premium (\$20000.00). The current applicant's profile is better, likely due to a lower overall risk profile (e.g., lower BMI, better lifestyle, or occupation) than POL-25755, resulting in a better risk class and lower premium.

* **Application POL-31852 (Similarity: 38.8%):** This case had the highest Risk Class (7) and Premium

(\$30000.00) among the similar applications. The current applicant's risk factors are clearly much more favorable than POL-31852, justifying a significantly lower risk class and premium.

The current application's Risk Class 3 and corresponding premium of \$14,000.00 are consistent with being slightly riskier than a Class 2 client but significantly less risky than Class 5 or 7 clients, as evidenced by the historical data. The machine learning model appears to position the applicant appropriately relative to the provided benchmarks.

3. Anomaly & Fraud Detection

* **Elevated BMI:** The most prominent anomaly is the BMI of 32.39, which indicates obesity. While the applicant reports exercising and no alcohol consumption, the high BMI remains a significant health concern that warrants further investigation, particularly for health insurance. It's not necessarily an inconsistency but a key risk driver.

* **Coverage Amount vs. Income:** A coverage amount of \$650,000.00 for Health insurance is substantial, especially compared to an annual income of \$50,000. While not directly indicative of fraud for health insurance (as medical costs can be very high irrespective of income), it could suggest a desire for comprehensive coverage for anticipated high medical expenses, possibly linked to an undisclosed health concern or the high BMI. This warrants further scrutiny to ensure the coverage is justified and affordable for the client over time.

* **Prediction Confidence:** The confidence score of 55.5% is relatively low. This suggests the model might have some uncertainty in classifying this specific profile, possibly due to the conflicting factors (high BMI vs. positive lifestyle) or a lack of highly similar cases in its training data. This low confidence is a flag for potential manual review.

4. Premium Recommendation

The calculated annual premium of \$14,000.00 appears to be reasonably justified:

* **Risk Class 3:** This class indicates a moderate-to-above-average risk profile.

* **Comparative Premium:** It is higher than the \$12,000.00 premium for Risk Class 2 (POL-2626), which is expected given the higher predicted risk class of the current applicant. It is significantly lower than the premiums for Risk Class 5 (\$20,000.00) and Risk Class 7 (\$30,000.00), aligning with the current applicant's better risk classification than those cases.

* **Balancing Factors:** The premium reflects the elevated risk due to obesity (BMI 32.39) but is mitigated by the favorable lifestyle factors (non-smoker, exercise, no alcohol, no family history, no previous claims).

Given the factors, the premium of \$14,000.00 is a plausible actuarial estimate for a Risk Class 3 individual with this profile.

5. Underwriting Decision Recommendation

****Recommendation: Refer for Manual Review (with potential for approval at current premium, possibly with surcharges based on medical review)****

****Justification:****

While the ML classifier provides a predicted risk class and premium, the low prediction confidence (55.5%) combined with the critical health indicator of obesity (BMI 32.39) necessitates a deeper human review.

* **High BMI:** A BMI of 32.39 is a significant health risk factor for health insurance. A manual underwriter should request an Attending Physician's Statement (APS) or medical examination to assess the specific health implications of the obesity, any co-morbidities, and the overall health status despite reported exercise.

* **Coverage Amount:** The high coverage amount (\$650,000.00) for health insurance, relative to income and in the context of high BMI, warrants a review to confirm its necessity and the applicant's long-term ability to afford the premium.

* **Conflicting Factors:** The positive lifestyle factors (exercise, non-smoker, no alcohol) are favorable, but the high BMI is a strong counter-indicator. A manual review can reconcile these to determine the true overall health risk.

The applicant's profile has many positive aspects, so outright rejection is not recommended without further information. However, due to the identified anomalies and the moderate confidence of the ML model, a manual review is essential to accurately assess the risk and confirm the proposed premium or adjust it with a surcharge if underlying health issues related to BMI are discovered.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$14,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.